

The Explicit Transition System for Certified Joint Steering: Round-17, Implementing the Eleventh Review’s Work Order

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July 13, 2026

Abstract

The eleventh review — the first in this project to find no counterexample — withheld certification of the round-16 soundness theorem pending six repairs: an explicit finite transition system for the reachability fixed point (R1), an ordered-pair transport rule (R2), a proof that actual unfolding states are contained in the computed reachable sets (R3), the separator-covers-core invariant stated and used (R4), explicit pair orientation (R5), and stored witness choices (R6). This delta supplies all six. The transition system is defined by a three-case row decomposition (continued / parent-fresh / core) made total by the now-explicit normal form of gluings; singleton states carry a persistent core row; ordered pairs are seeded co-patterned, created at fresh birth, and transported with the singleton transitions while preserving their realized mutual value. The operational-inclusion invariant (R3) is proved by induction on the unfolding, with the shared-core cases discharged by core-row persistence and birth-pattern membership; a no-overwrite lemma underpins the transport step. The witness map makes the canonical unfolding a function of the certificate alone, so the determinism claim is now exact. An executable rendering of the transition system (WP30) machine-checks the inclusion invariant on 1,200 random unfoldings (no violations), shows transport is load-bearing (83/83 catalogues have transported pairs a birth-only fixed point misses), and confirms S4-validated steering yields closed unfoldings. Unchanged and open: the width budgets (F6) and uniformization (W2’). Status: *strongly supported, not certified*.

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*Round-17 delta on rounds 15/16 ([papers/fable5_round15/](#), [papers/fable5_round16/](#)), implementing repairs R1–R6 of the eleventh review ([papers/cold_review_round16/formal_referee_report_alci_rcc5.pdf](#)), whose ordered-pair transport rule (its Appendix B) is adopted here in substance. Everything not amended is inherited. Project conventions: eleven reviews — ten defects with witnesses, one no-counterexample verdict with required repairs; this delta is unreviewed and should be presumed to contain the next finding. Designated weak points: Section 9.

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1 Scope

This delta amends round 16’s Definition 2.3 (the prose fixed point), strengthens round 15’s (Q3)/(Q5) into the explicit normal form the eleventh review demanded, and adds the witness map. The findings map: F1/F3 → Sections 3–4; F2/F7 → Section 2; F4 → Section 5; F5 → ordered pairs throughout; F6 and W2’ remain open and are not touched (Section 6).

2 The normal form of gluings (R4; repairs F2/F7)

Definition 2.1 (Gluing normal form; condition (Q3’)). An unfolding attaches each new pattern copy k at a gluing g to a unique *parent* copy k' (the root copy attaches to the core pattern). The certificate’s gluing declaration for g lists the separator σ_g exhaustively, and validity requires:

- (N1) $V_k \cap U = \sigma_g$, where U is the set of occurrences existing before the step: every pre-existing member of the new copy is in the separator — there is no second channel by which an old occurrence can be co-patterned with fresh members;
- (N2) $\sigma_g \subseteq V_{k'} \cup \text{core}$: separator members are members of the parent copy or core occurrences;
- (N3) $\text{core} \subseteq \sigma_g$ for every gluing g : core references are permanent interface members, so every pattern copy contains the core and core members are continued at every step.

Remark 2.2 (The ambient dichotomy). By (N1), at every step each existing occurrence is either in σ_g — co-patterned with the fresh members, receiving *pattern* values — or outside V_k entirely, receiving *steering* values. The eleventh review’s F2 failure mode (an old core occurrence co-patterned with fresh members yet outside the separator, creating two competing value sources) is excluded by definition, not by argument. The review’s own observation is retained as a check: for $s \in \sigma_g$, the row entry of s at itself is EQ, and $\text{Comp}(\text{EQ}, N_k(s, b)) = \{N_k(s, b)\}$, so even a redundant steering application to separator members would be forced to the pattern value.

3 The explicit transition system (R1, R2, R5; repairs F1/F3/F5)

Fix a certificate with catalogue, gluing declarations in normal form, and steering functions f_g .

Definition 3.1 (Singleton states). The state of an existing occurrence y at a gluing g is

$$\text{St}_g(y) = (\text{tp}(y), \rho \upharpoonright (\{y\} \times \text{core}), \rho \upharpoonright (\{y\} \times \sigma_g), (\text{Safe}(\text{tp } y, \tau_k b))_{b \in B_g}),$$

its type, *persistent core row*, separator row, and safety signature. The core row is set at birth — by (N3) the core is contained in y ’s birth pattern, so $\rho(y, c)$ is an in-pattern value for every core member c — and never changes thereafter (core members are continued at every step, and values are never overwritten: Lemma 3.4).

Definition 3.2 (Singleton transitions). Let g' be a child gluing of the copy k attached at g (so $\sigma_{g'} \subseteq V_k \cup \text{core}$ by (N2)). For an occurrence y ambient at g (i.e. $y \notin V_k$), the transition $T_{g \rightarrow g'}$ sends $\text{St}_g(y)$ to the state whose separator row is given by the *three-case decomposition*: for $s \in \sigma_{g'}$,

$$\rho(y, s) = \begin{cases} (\text{continued}) & s \in \sigma_g : \text{ the } \sigma_g\text{-row entry of } y; \\ (\text{parent-fresh}) & s \in B_g : f_g(\text{St}_g(y))(s); \\ (\text{core}) & s \in \text{core} : \text{ the core-row entry of } y. \end{cases}$$

By (N2)+(N3) these cases are exhaustive, so $T_{g \rightarrow g'}$ is a total function on states; the type, core row, and (certificate-determined) safety signature components update trivially. For $y \in V_k$ (a member of the parent copy, including its fresh members), the state at g' is read off the pattern network and, for y 's relations to $\sigma_{g'}$ -members, consists of in-pattern values only — these are the *birth-adjacent* states, computable from the catalogue.

Definition 3.3 (Ordered pair states; the transport rule). Pair states are *ordered* triples $(\varsigma, \varsigma', W)$ with $W \in \text{Rel} \setminus \{\text{EQ}\}$; the converse triple $(\varsigma', \varsigma, W^{-1})$ is always added simultaneously. The reachable pair set is the least set closed under:

- (P1) **Co-patterned seeds**: for any two members a, b of one pattern copy at its birth gluing, $(\text{St}(a), \text{St}(b), N_k(a, b))$.
- (P2) **Creation**: when a fresh b is created at g and y is ambient, $(\text{St}_g(y), \text{St}_g(b), f_g(\text{St}_g(y))(b))$, where $\text{St}_g(b)$ is b 's birth state.
- (P3) **Transport** (the eleventh review's Appendix-B rule): if $(\varsigma, \varsigma', W)$ is reachable before crossing a child gluing g' and both singleton transitions are defined, then $(T_{g \rightarrow g'}(\varsigma), T_{g \rightarrow g'}(\varsigma'), W)$ is reachable after it. The mutual value W is preserved — justified by Lemma 3.4.

All components range over finite alphabets (types; rows over core and over separators of bounded size; signatures), so the least fixed point is finite and effectively computable. Condition (Q4') of round 16 — in particular (S4) — is checked on exactly these sets.

Lemma 3.4 (No overwrite). *In an unfolding satisfying (Q3'), every pair of occurrences receives its value exactly once: in-pattern at the first gluing where the two are co-patterned, or by steering at the birth step of the younger. Later steps never reassign it.*

Proof. At a step attaching copy k : values assigned are $\sigma_g \times B_g$ and $B_g \times B_g$ (pattern values on fresh pairs) and $(U \setminus \sigma_g) \times B_g$ (steering). By (N1), no pair of pre-existing occurrences is touched: pattern values on $\sigma_g \times \sigma_g$ pre-exist, and gluing coherence (round-15 (Q3)) requires the declared pattern network to agree with them, so no reassignment occurs there either. Every touched pair involves a fresh member, hence had no prior value. $\square$

4 The operational-inclusion invariant (R3; repairs F1/F3)

Theorem 4.1 (Inclusion). *In every (finite prefix of an) unfolding of a certificate in normal form: at every gluing g , the state $\text{St}_g(y)$ of every existing occurrence lies in the computed reachable singleton set, and for every ordered pair (y, z) of existing occurrences, $(\text{St}_g(y), \text{St}_g(z), \rho(y, z))$ lies in the computed reachable pair set.*

Proof. Induction on the number of steps. *Base:* the root copy attached to the core pattern. Every occurrence is a member of one of the two patterns; all states are birth(-adjacent) states, seeded; all pairs are co-patterned, seeded by (P1) — including core-core pairs (co-patterned in the core pattern) and core-root pairs (co-patterned in the root copy by (N3)).

Step: suppose the invariant holds before attaching copy k at g , and consider the next gluing g' (a child of k , or a later gluing elsewhere in the tree order; the argument is the same with T composed along the corresponding path). Three obligations:

(a) *Ambient singleton states update by $T_{g \rightarrow g'}$.* For y ambient at g , its $\sigma_{g'}$ -row is given pointwise by exactly the three cases of Definition 3.2: $s \in \sigma_g$ gives a continued value (unchanged by Lemma 3.4); $s \in B_g$ gives the value assigned at step g , which was $f_g(\text{St}_g(y))(s)$ by the construction — here (N1) is used: $y \notin V_k$, so the value was a steering value, not a pattern value; $s \in \text{core}$ gives the persistent core-row entry. By (N2)+(N3) there is no fourth kind of separator member. Hence $\text{St}_{g'}(y) = T_{g \rightarrow g'}(\text{St}_g(y))$, which is in the computed set by induction and closure under transitions.

(b) *Fresh states are seeds.* Members of k have birth(-adjacent) states computable from the catalogue (Definition 3.2), seeded.

(c) *Pairs.* Pairs among pre-existing occurrences: their mutual value is unchanged (Lemma 3.4) and both endpoint states update by T , so the pair is covered by transport (P3) applied to the inductively-reachable pair. Pairs of one ambient and one fresh occurrence: covered by creation (P2) with the assigned steering value. Pairs among members of k (fresh-fresh, fresh-separator, separator-separator): co-patterned seeds (P1), with the separator-separator case agreeing with pre-existing values by gluing coherence. Core cases are instances of the above, not exceptions: core-core and core-fresh pairs are co-patterned at every step by (N3); core-ambient pairs were born co-patterned at the ambient occurrence's birth (again (N3)) and are transported thereafter, their W persistent by Lemma 3.4 and the core row persistent by Definition 3.1. $\square$

Corollary 4.2. *Round-16 Theorem 3.2 (deterministic canonical completion) holds with its Definition 2.3 replaced by Definitions 3.1–3.3: at every step, the old-old-fresh triangle class is discharged by (S4) checked on a pair that Theorem 4.1 guarantees is in the certified set with its current states and realized mutual value; the other three classes are unchanged. Consequently the certificate-to-model soundness theorem (round-16 Corollary 3.3) stands on the explicit system.*

5 The witness map (R6; repairs F4)

Definition 5.1 (Condition (Q2')). The certificate carries a total map w assigning to every pattern k , member $a \in V_k$, and demand $\exists R.D \in \tau_k(a)$ either an in-pattern witness $b \in V_k$ with $N_k(a, b) = R$, $D \in \tau_k(b)$, or a designated gluing edge of $\mathcal{G}$ at which a is a separator member of an adjacent pattern fulfilling the demand in-pattern.

With w stored, the canonical unfolding is generated deterministically from the root: each step's attached copies are those designated by w for the unfulfilled demands of the frontier. The round-16 claim is restated exactly: *the unfolding and its completion are functions of the certificate*; no choice is made outside the certificate. (The eleventh review classified the previous formulation as a presentation-level defect; nothing in the soundness argument changes.)

6 What remains open

Unchanged by this delta, and explicitly still open: **F6** — the width recurrence's non-iteration budgets for boundary and stratum insertions (round-15 Section 6) remain assertions awaiting a

derivation from the saturation rules; **W2'** — uniformization (round-16 Section 4), with its measured 98.6% coarse-granularity rate and two sketched refinement routes. Both sit on the completeness side; the eleventh review certified (F8) that soundness does not depend on W2', and this delta introduces no new dependence.

7 Machine checks (WP30)

`verification/python/wp30_round17_transition_system.py` implements the transition system executably (synthetic level: rows, cores, transport, orientation; type and safety components ride along identically) and checks, all PASS: **A** (R3): on 120 random catalogues with shared cores, 10 random unfoldings each (1,200 simulations): every actual singleton state and every actual ordered pair, with its realized mutual value, lies in the computed reachable sets — zero containment failures. **B** (R2 is load-bearing): recomputing the fixed point *without* transport leaves actual transported pairs uncovered in 83/83 catalogues that have them — the eleventh review's F1, confirmed at scale. **C**: when the canonical steering satisfies (S4) on the computed pairs, all simulated unfoldings are closed atomic frames (100 catalogues, zero failures).

8 Response map

R1	Definitions 3.1–3.3: explicit finite transition system, singleton and ordered pair states separated.
R2	(P3), adopted from the review's Appendix B; justified by Lemma 3.4.
R3	Theorem 4.1, with the core cases discharged by (N3), core-row persistence, and no-overwrite; machine-checked (WP30 A/B).
R4	(Q3') = (N1)–(N3), Section 2; used, not just stated, in Definition 3.2 (exhaustiveness of the three-case decomposition) and Theorem 4.1(a).
R5	Ordered triples with simultaneous converses, Definition 3.3.
R6	(Q2'), Section 5; determinism restated exactly.

9 What would invalidate this delta

- (Y1) An unfolding shape violating the gluing normal form that the extraction direction genuinely needs — i.e. a satisfiable concept whose Theorem-A presentations cannot be arranged with $V_k \cap U = \sigma_g$, $\sigma_g \subseteq V_{k'} \cup \text{core}$, $\text{core} \subseteq \sigma_g$. (N2) in particular constrains separators to parent-plus-core; if saturation ever requires sharing with a non-parent, non-core copy, the transition system's three-case decomposition is incomplete.
- (Y2) A failure of Lemma 3.4 under some declaration pattern (e.g. overlapping gluings touching the same pair twice with different values) — excluded by (N1)+(Q3) as argued, but the arguing is new.
- (Y3) The standing open items: F6 (width budgets), W2' (uniformization), and the single external input (RCC5 patchwork).

10 Status

Eleven reviews: ten defects with witnesses, one no-counterexample verdict with required repairs — now implemented. By the reviewer’s own criterion (“with these changes, I would regard the intended soundness argument as very likely correct”), the soundness half is, for the first time, fully written: explicit system, proved inclusion, no free choices, machine-checked at every finite joint. It remains *unreviewed in this form*, and the completeness side retains its two named open lemmas. The decidability theorem for $\mathcal{ALCI}_{\text{RCC5}}$ remains **strongly supported, not certified**. The natural next steps: a verification pass on this delta (twelfth review), and — since Definitions 3.1–3.3 are already in transition-system form — the start of the Lean mechanization of Theorem 4.1 and round-16 Theorem 3.2, which are now transcription rather than research.

References

- [1] Fresh GPT-5.5 session. *Adversarial Referee Report on the Cluster-Quasimodel Decision Layer for $\mathcal{ALCI}_{\text{RCC5}}$* , July 13, 2026. [papers/cold.review_round16/](#).
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- [4] J. Renz and B. Nebel. On the complexity of qualitative spatial reasoning. *Artificial Intelligence* 108 (1999).